

《数据库系统原理》实验报告（）					
题目：DDL 语言实验					
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实验环境： 硬件环境：windows\wsl, Docker Desktop 版本 28.0.1 软件环境：OceanBase 社区版镜像:oceanbase/oceanbase-ce:lastest					
实验步骤及结果截图： 1. 建立 table <pre>create table books(No Integer not null, Title Varchar(50) not null, Author Varchar(50) not null, publication_year Integer not null, C_no Integer not null, primary key(No), foreign key(C_no) references category(No)) CREATE TABLE category (No INT NOT NULL PRIMARY KEY, Name VARCHAR(50) NOT NULL); CREATE TABLE readers(No INT NOT NULL PRIMARY KEY, Name VARCHAR(20) NOT NULL, Gender VARCHAR(10) NOT NULL CHECK (Gender IN ('Female', 'Male')), Tel VARCHAR(20) NOT NULL); CREATE TABLE borrow_records (No INT NOT NULL PRIMARY KEY, B_no INT NOT NULL, R_no INT NOT NULL, Borrow_date DATE NOT NULL, Return_date DATE, FOREIGN KEY (B_no) REFERENCES books(No), FOREIGN KEY (R_no) REFERENCES readers(No));</pre> 2. 插入表单项 2.1 Category 表插入 <pre>obclient(root@sys)[mamba]> INSERT INTO category (No,Name) VALUES -> (1,'计算机'), -> (2,'文学'), -> (3,'历史'), -> (4,'科学');</pre> 2.2 Books 表的插入					

```
obclient(root@sys)[mamba]> INSERT INTO books(No, Title, Author, publication_year, C_no) VALUES
-> (1, 'Python编程实战', '张三', 2021, 1),
-> (2, '百年孤独', '加西亚·马尔克斯', 1967, 2),
-> (3, '明朝那些事儿', '当年明月', 2006, 3),
-> (4, '时间简史', '霍金', 1988, 4),
-> (5, 'Java核心技术', '凯S.霍斯特曼', 2019, 1);
```

2.3 readers 表的插入

```
obclient(root@sys)[mamba]> INSERT INTO readers (No, Name, Gender, Tel) VALUES
-> (1, '小明', 'Male', '13800138000'),
-> (2, '小红', 'Female', '13900139000'),
-> (3, '小刚', 'Male', '13600136000'),
-> (4, '小美', 'Female', '13700137000'),
-> (5, '小强', 'Male', '13300133000');
```

2.4 borrow_records 表的插入

```
obclient(root@sys)[mamba]> INSERT INTO borrow_records (No, B_no, R_no, Borrow_date, Return_date)
-> VALUES
-> (1,1,1, '2024-10-01', '2024-10-20'),
-> (2,2,2, '2024-10-05', '2024-10-25'),
-> (3,1,3, '2024-10-10', '2024-10-30'),
-> (4,3,4, '2024-10-15', '2024-11-01'),
-> (5,2,5, '2024-10-20', NULL);
```

3.使用比较运算符查询 2020 年以后出版的图书的 Title 和 publication_year。

[172.17.0.2:2882] [2025-04-15 10:51:08.617648] [YB42AC110002-000632CEDDBA8F4E-0-0]

```
obclient(root@sys)[mamba]> SELECT title, publication_year
-> FROM books
-> WHERE publication_year > 2020;
```

```
+-----+-----+
| title           | publication_year |
+-----+-----+
| Python编程实战 |          2021   |
+-----+-----+
1 row in set (0.003 sec)
```

4. 查询借阅过编号为 3 的图书的读者的 Name 和 Tel。

```
obclient(root@sys)[mamba]> SELECT r.name, r.tel
-> FROM readers r
-> JOIN borrow_records br ON r.No = br.R_no
-> WHERE br.B_no = 3;
```

```
+-----+-----+
| name | tel       |
+-----+-----+
| 小美 | 13700137000 |
+-----+-----+
1 row in set (0.023 sec)
```

5.查询作者姓名中包含“张”字的图书信息。

```
obclient(root@sys)[mamba]> SELECT No,Title, Author, publication_year
-> FROM books
-> WHERE Author like '%张%';
+-----+-----+-----+-----+
| No   | Title           | Author | publication_year |
+-----+-----+-----+-----+
| 1    | Python编程实战 | 张三   | 2021             |
+-----+-----+-----+-----+
1 row in set (0.002 sec)
```

6.查询所有男性读者的借阅记录，包括读者姓名、借阅图书 Title、借阅日期和归还日期，结果按借阅日期升序排列。

```
obclient(root@sys)[mamba]> SELECT r.Name,b.Title,br.Borrow_date,br.Return_date
-> FROM readers r
-> JOIN borrow_records br ON r.No = br.R_no
-> JOIN books b ON b.No = br.B_no
-> WHERE r.Gender = 'Male'
-> ORDER BY br.Borrow_date ASC;
+-----+-----+-----+-----+
| Name  | Title           | Borrow_date | Return_date |
+-----+-----+-----+-----+
| 小明  | Python编程实战 | 2024-10-01  | 2024-10-20  |
| 小刚  | Python编程实战 | 2024-10-10  | 2024-10-30  |
| 小强  | 百年孤独       | 2024-10-20  | NULL        |
+-----+-----+-----+-----+
3 rows in set (0.008 sec)
```

7.查询女性读者的总人数。

```
obclient(root@sys)[mamba]> SELECT COUNT(*) FROM Readers
-> WHERE Gender = 'Female';
+-----+
| COUNT(*) |
+-----+
| 2        |
+-----+
1 row in set (0.002 sec)
```

8. 查询借阅时长超过 15 天的借阅记录，包括读者姓名、图书 Title。

```
obclient(root@sys)[mamba]> SELECT r.Name,b.Title FROM
-> borrow_records br
-> JOIN readers r ON r.No = br.R_no
-> JOIN books b on b.No = br.B_no
-> WHERE br.Return_date - br.Borrow_date > 15;
+-----+-----+
| Name  | Title           |
+-----+-----+
| 小明  | Python编程实战 |
| 小刚  | Python编程实战 |
| 小红  | 百年孤独       |
| 小美  | 明朝那些事儿   |
+-----+-----+
4 rows in set (0.007 sec)
```

9.向 borrow_records 表中插入一条记录（借阅编号为 6，图书编号为 5，读者编号为 5，借阅日期为 '2025-03-24'，归还日期为空）。

```
obclient(root@sys)[mamba]> INSERT INTO borrow_records(No,B_no,R_no,Borrow_date)
-> VALUES(6,5,5,'2025-03-24');
Query OK, 1 row affected (0.031 sec)
```

```
obclient(root@sys)[mamba]> Select * From borrow_records;
+-----+
| No | B_no | R_no | Borrow_date | Return_date |
+-----+
| 1 | 1 | 1 | 2024-10-01 | 2024-10-20 |
| 2 | 2 | 2 | 2024-10-05 | 2024-10-25 |
| 3 | 1 | 3 | 2024-10-10 | 2024-10-30 |
| 4 | 3 | 4 | 2024-10-15 | 2024-11-01 |
| 5 | 2 | 5 | 2024-10-20 | NULL |
| 6 | 5 | 5 | 2025-03-24 | NULL |
+-----+
6 rows in set (0.001 sec)
```

10. 计算每本图书的平均借阅时长，并输出平均借阅时长超过 15 天的图书的 No。

```
obclient(root@sys)[mamba]> SELECT B_no
-> FROM borrow_records
-> GROUP BY B_no
-> HAVING AVG(DATEDIFF(Return_date,Borrow_date)) > 15;
+-----+
| B_no |
+-----+
| 1 |
| 2 |
| 3 |
+-----+
3 rows in set (0.004 sec)
```

11. 查询借阅过编号为 1 或者编号为 2 图书的读者 No。

```
obclient(root@sys)[mamba]> SELECT r.No
-> FROM readers r
-> JOIN borrow_records br ON r.No = br.R_no
-> WHERE br.B_no IN (1,2);
+-----+
| No |
+-----+
| 1 |
| 2 |
| 3 |
| 5 |
+-----+
4 rows in set (0.003 sec)
```

12. 查询既借阅过编号为 1 又借阅过编号为 3 图书的读者 No。

```
obclient(root@sys)[mamba]> SELECT r.No
-> FROM readers r
-> JOIN borrow_records br ON r.No = br.R_no
-> WHERE br.B_no = 1
->
-> INTERSECT
->
-> SELECT r.No
-> FROM readers r
-> JOIN borrow_records br ON r.No = br.R_no
-> WHERE br.B_no = 3;
Empty set (0.016 sec)
```

13. 建立一个包含图书名字，作者，出版年份和图书类别的视图（赋予列名为 stitle, sauthor, spublicationyear, categoryname）。【create view】

```
obclient(root@sys)[mamba]> CREATE VIEW myview AS
-> SELECT
-> b.Title AS stitle,
-> b.Author AS sauthor,
-> b.publication_year AS spublicationyear,
-> c.Name AS categoryname
-> FROM books b
-> JOIN category c ON b.C_no = c.No;
Query OK, 0 rows affected (0.090 sec)
```

```
obclient(root@sys)[mamba]> SELECT * FROM myview;
+-----+-----+-----+-----+
| stitle          | sauthor          | spublicationyear | categoryname |
+-----+-----+-----+-----+
| Python编程实战 | 张三              | 2021 | 计算机 |
| 百年孤独        | 加西亚·马尔克斯 | 1967 | 文学   |
| 明朝那些事儿    | 当年明月          | 2006 | 历史   |
| 时间简史        | 霍金              | 1988 | 科学   |
| Java核心技术    | 凯S.霍斯特曼     | 2019 | 计算机 |
+-----+-----+-----+-----+
5 rows in set (0.015 sec)
```

出现的问题:

1. 插入数据重复导致 PRIMARY KEY 报错

```
Database changed
obclient(root@sys)[mamba]> INSERT INTO category (No,Name) VALUES
-> (1,'计算机'),
-> (2,'文学'),
-> (3,'历史'),
-> (4,'科学');
ERROR 1062 (23000): Duplicate entry '1' for key 'PRIMARY'
```

2. 删除表格报错，由于别的表格有外键约束

```
obclient(root@sys)[mamba]> drop table readers;
ERROR 3730 (HY000): Cannot drop table 'readers' referenced by a foreign key constraint 'borrow_records_OBFK_1744706812420152' on table 'borrow_records'
[172.17.0.2:2882] [2025-04-15 11:40:31.845488] [YB42AC110002-000632CEDDBA8F7A-0-0]
```

解决方案:

1. 检查插入数据，剔除重复插入数据

2. 禁用外键约束检查，然后删除表后再重新启用外键约束检查

```
[172.17.0.2:2882] [2025-04-15 11:40:31.845488] [YB42AC110002-000632CEDDBA8F7A-0-0]
```

```
obclient(root@sys)[mamba]> SET FOREIGN_KEY_CHECKS = 0;
```

```
obclient(root@sys)[mamba]> SET FOREIGN_KEY_CHECKS = 1;
```